

Heathcare Data Analysis Report

Muhammad Ali

October 26, 2025

Introduction:

This dataset presents a comprehensive collection of hospital patient record metadata, capturing core details such as patient name, gender, blood type, insurance information, medication, and more. In addition, it also provides identifiers such as the doctor's name, hospital, billing amount, test results and other attributes that are relevant for a health-care environment. Utilizing the metadata, this will enable analysis beyond the surface level revealing correlations between the identifiers and hospital operations when it comes to patients.

The central focus of this project is to identify and interpret patterns by using Python, SQL, and the visualization tool Power BI to answer questions such as:

- How do demographic factors such as gender and blood type relate to common treatments, lengths of stay, or medication regimens?
- Does a certain hospital have a different time length between admission and discharge date?
- Are there trends between blood types with certain medical conditions?

Through addressing these questions and future questions, this analysis seeks to deliver actionable insights into the functioning and optimization of patient care, administrative workflow, and whatever findings that are hoped to be unveiled through this analysis.

Ultimately, this report aims to elevate the understanding of hospital operations by transforming raw metadata into clear, actionable intelligence about patient needs, care quality, and institutional performance, fueling continuous improvement and innovative practice across the healthcare ecosystem.